

THE THREE CADENCES OF EMBODIED INTELLIGENCE

System 2 Deliberation (1 Hz) ↔ System 1.5 Planning (20 Hz) ↔ System 1 Reflex (1000 Hz)

CADENCE 1: SYSTEM 2

Deliberate Reasoning (1–2 Hz)

Period: 500–1000 ms · Host MPU / VLM

- Semantic task grounding
- Coarse 3D target bounding
- Expiring intent leases t_{expire}
- High latency tolerance



CADENCE 2: SYSTEM 1.5

Trajectory Planning (20–50 Hz)

Period: 20–50 ms · Host MPU / NPU

- Diffusion policy / ACT chunks
- $H=16$ step horizon generation
- C^2 continuous spline fitting
- Temporal overlap ensembling



CADENCE 3: SYSTEM 1

Safety Reflex (1000 Hz)

Period: 1.0 ms · Real-Time MCU

- Active-Set CBF QP solver
- Control Barrier Invariant $h(x) \geq 0$
- Dynamic stopping envelope check
- Deterministic FreeRTOS priority